CAREERS 360

PREPARATION Series

NEET UG 2027

Genetics and Evolution PYQs

100+ Previous Year Questions with Detailed Solutions

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Genetics and Evolution Previous Year Questions

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About this eBook

Dear NEET Aspirants,

Welcome to your ultimate **Year-wise NEET PYQs Guide for Genetics and Evolution!** This eBook is designed to help you master one of the most crucial sections of Biology, which holds significant weight in the NEET exam. By focusing on questions from **previous years (PYQs) organised year-wise**, this eBook ensures that you understand the exam pattern, frequently tested topics, and question trends over the years.

What does this eBook contain?

Year-wise NEET PYQs Compilation – We have systematically arranged previous years' questions year-wise to help you analyze trends and understand how concepts are tested across different NEET exams.

Detailed Solutions – Every question comes with a step-by-step explanation to ensure conceptual clarity, reinforcing the fundamentals of Genetics and Evolution.

Topic-Wise Yearly Breakdown – The eBook covers the following topics:

Mendelian Genetics – Year-wise PYQs on Mendel's laws, monohybrid & dihybrid crosses, and deviations from Mendelian ratios.

Molecular Basis of Inheritance – NEET questions on DNA structure, replication, transcription, translation, genetic code, and mutations.

Chromosomal Basis of Inheritance – PYQs on linkage, recombination, chromosomal disorders, and sex determination.

Evolutionary Biology – Year-wise questions on Darwinism, speciation, adaptive radiation, and Hardy-Weinberg principle.

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Biotechnology and Genetic Engineering – PYQs covering rDNA technology, gene cloning, PCR, and applications of biotechnology.

How Will This eBook Benefit You?

NEET-Oriented Preparation – By focusing on **PYQs**, you'll develop a deep understanding of how questions are framed and what to expect in the exam.

Higher Accuracy & Speed – Practicing past questions improves your problem-solving speed and accuracy, giving you an edge over others.

Boosts Confidence – Since these questions have appeared in previous **NEET exams**, solving them will **strengthen your confidence and enhance your exam strategy**.

Smart & Efficient Learning – Instead of studying randomly, this eBook ensures **structured, high-yield learning**, helping you focus on scoring topics without unnecessary distractions.

Mastering **Genetics and Evolution PYQs** will give you the competitive advantage needed to excel in NEET. Study smart, stay consistent, and take one step closer to your dream of becoming a medical professional!

Importance Of Genetics and Evolution In NEET

The unit Genetics and Evolution is crucial for NEET aspirants as it covers key concepts like Mendelian laws, genetic variations, and evolutionary mechanisms, which are essential for understanding how traits are inherited and how species evolve. This unit is divided into several chapters: Principles of Inheritance, which focuses on the fundamental laws and theories of genetic inheritance, such as Mendel's laws and chromosomal disorders; and Evolutionary Processes, which explores how genetic variations lead to evolutionary changes through mechanisms like mutation, gene flow, and natural selection. Mastering these concepts is vital for scoring well in NEET, as they form the basis for careers in genetics research and related fields.

Weightage Of Genetics and Evolution In NEET

Genetics and Evolution hold significant weightage in NEET, comprising approximately 18% of the marks in the Biology section. Typically, around 15 questions are asked from this chapter, carrying a total of 60 marks. This substantial weightage makes Genetics and Evolution a critical chapter for students aiming to score high marks in NEET. The consistent appearance of questions from this chapter in previous years underscores its importance for the exam. Mastering concepts like Mendelian laws, genetic variations, and evolutionary mechanisms is essential for success in NEET.

Principles of Inheritance and Variation

The chapter **"Principles of Inheritance and Variation"** in biology discusses how traits are passed from one generation to the next and how variations arise.

Here is a table summarizing the key concepts covered in the chapter "Principles of Inheritance and Variation":

Topic	Description
Mendelian Inheritance	Explains how traits are passed from parents to offspring using Mendel's Laws: Law of Dominance, Law of Segregation, and Law of Independent Assortment.
Incomplete Dominance	A phenotype is a blend of the two parental traits, neither completely dominant nor recessive.
Co-Dominance	Both alleles have an equal effect on the phenotype, e.g., ABO blood groups.
Multiple Alleles	More than two alleles for a gene, e.g., ABO blood groups.
Pleiotropy	One gene affects multiple traits.
Polygenic Inheritance	Traits are influenced by multiple genes, e.g., skin colour and height.

Chromosomal Theory Of Inheritance	Explains how chromosomes carry genes and influence inheritance.
Linkage And Recombination	Genes on the same chromosome may be linked, but recombination can occur during meiosis.
Sex Determination	Mechanisms by which sex is determined in different species.
Mutations	Changes in the DNA sequence can affect traits.
Genetic Disorders	Conditions caused by genetic mutations, e.g., haemophiliaanaemiaa and sickle cell anaemia.

NEET 2026

Ques: Which of the following statements are true with reference to sex determination in honeybees?
A. An offspring formed from the union of a sperm and an egg develops as a female (queen or worker).
B. An unfertilized egg develops as a male by parthenogenesis.
C. A male has half the number of chromosomes as a female.
D. Males produce sperm by meiosis.
E. Honeybees have a haplodiploid sex-determination system.
Choose the correct answer from the options given below:

Option 1) A, B, C and E only

Option 2) A, B, C and D only

Option 3) B, C, D and E only

Option 4) A, B, D and E only

Solution - In honeybees, fertilised eggs (2n) form females, unfertilised eggs (n) form males by parthenogenesis. Males are haploid and produce sperm by mitosis, not meiosis, under this haplodiploid sex-determination system. Statement D is false because haploid males cannot undergo meiosis to produce gametes.

Hence, the correct answer is option 1) A, B, C and E only.

Ques: Which one of the following disorders is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule?

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Option 1) Phenylketonuria

Option 2) Haemophilia

Option 3) Sickle-cell anaemia

Option 4) Thalassemia

Solution - Sickle-cell anaemia is caused by a point mutation where Glu is substituted by Val at position 6 of the beta-globin chain. This changes the hydrophilic glutamic acid to hydrophobic valine, altering haemoglobin structure so that red blood cells become sickle-shaped under low-oxygen conditions.

Hence, the correct answer is option 3) Sickle-cell anaemia.

Ques: Match List I with List II:
List I (Genetic phenomenon) - List II (Example)
A. Incomplete dominance - I. Human skin colour
B. Co-dominance - II. Inheritance of flower colour in Antirrhinum sp.
C. Pleiotropy - III. Phenylketonuria disease in humans
D. Polygenic inheritance - IV. ABO blood groups
Choose the correct answer from the options given below:

Option 1) A-I, B-IV, C-III, D-II

Option 2) A-I, B-III, C-II, D-IV

Option 3) A-II, B-IV, C-III, D-I

Option 4) A-II, B-I, C-III, D-IV

Solution - Incomplete dominance is seen in the inheritance of flower colour in Antirrhinum (A-II); co-dominance is seen in ABO blood groups (B-IV); pleiotropy is seen in phenylketonuria, where a single gene affects multiple traits (C-III); polygenic inheritance governs continuous variation such as human skin colour (D-I).

Hence, the correct answer is option 3) A-II, B-IV, C-III, D-I.

Ques: What is the probability of having children with 'O' blood group, where both mother and father are heterozygous for 'A' and 'B' blood group, respectively?

Option 1) 0%

Option 2) 50%

Option 3) 25%

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Option 4) 75%

Solution - Cross: IAi x IBi gives offspring genotypes IAIB, IAi, IBi and ii in equal proportion. Only the ii genotype gives blood group O, so the probability is 1/4 = 25%.

Hence, the correct answer is option 3) 25%.

Ques: In which animal do haploid cells divide mitotically to produce gametes?

Option 1) Male honeybees

Option 2) Male grasshoppers

Option 3) Male earthworms

Option 4) Male frogs

Solution - Male honeybees (drones) develop from unfertilised eggs and are therefore already haploid. Since meiosis reduces chromosome number by half, it is unnecessary in drones, and gamete formation instead occurs by mitosis.

Hence, the correct answer is option 1) Male honeybees.

Ques: The sixth mutant codon of beta globin gene causing polymerization of Haemoglobin and change in RBC shape is -----

Option 1) CAG

Option 2) AUG

Option 3) GUG

Option 4) GAG

Solution - In sickle-cell anaemia, the sixth codon of the beta-globin gene mutates from GAG to GUG. This substitution results in the production of Haemoglobin S, which polymerises at low oxygen levels and causes sickle-shaped RBCs.

Hence, the correct answer is option 3) GUG.

Ques: In a population of a grasshopper species, the chromosome number of some members is 23 and some other members possess 24 chromosomes. The 23 and 24 chromosome-bearing members in this species are -----.

Option 1) females and males, respectively

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Option 2) males and females, respectively

Option 3) all males

Option 4) all females

Solution - Grasshoppers show the XO type of sex determination. Females have two X chromosomes (24 total), while males have only one X chromosome (23 total, XO), both sexes having the same number of autosomes.

Hence, the correct answer is option 2) males and females, respectively.

Re-NEET 2026

Ques: Given below are two statements: one is labelled as Assertion A, and the other is labelled as Reason R.
Assertion A: In an experiment, Mendel observed that the F1 progeny plants are all tall and none are dwarfs.
Reason R: Stem height is a contrasting trait, with tall being dominant and dwarf being recessive.
In light of the above statements, choose the most appropriate answer from the options given below:

Option 1) A is not correct but R is correct

Option 2) Both A and R are correct and R is the correct explanation of A

Option 3) Both A and R are correct but R is not the correct explanation of A

Option 4) A is correct but R is not correct

Solution - Tallness is dominant over dwarfness; therefore, all F1 progeny obtained from a cross between pure tall and pure dwarf plants are tall, and R correctly explains why A is observed.

Hence, the correct answer is option 2) Both A and R are correct and R is the correct explanation of A.

Ques: For a person with blood group 'O', which of the following is not a possible combination of parents' blood group genotypes?

Option 1) Father: IAIB and Mother: IAi

Option 2) Father: IAi and Mother: Ii

Option 3) Father: IAi and Mother: IAi

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Option 4) Father: IBi and Mother: Ii

Solution - For a child to have blood group O, its genotype must be ii, requiring one i allele from each parent. A father with genotype IAIB carries no i allele at all, so he can never pass an i allele to his child, making an O blood group child impossible from this combination. All other listed combinations have at least one i allele in both parents, so an O-group child remains possible.

Hence, the correct answer is option 1) Father: IAIB and Mother: IAi.

Ques: Given below are two statements:
Statement I: Down's syndrome is caused by the absence of one of the X-chromosomes.
Statement II: Turner's syndrome is caused by the presence of an additional copy of the chromosomes.
In the light of the above statements, choose the correct answer from the options given below:

Option 1) Statement I is incorrect but Statement II is correct

Option 2) Both Statement I and Statement II are correct

Option 3) Both Statement I and Statement II are incorrect

Option 4) Statement I is correct but Statement II is incorrect

Solution - Down's syndrome is caused by trisomy of chromosome 21 (an extra chromosome), not the absence of an X chromosome. Turner's syndrome is caused by the absence of one X chromosome (45, XO), not by an additional chromosome copy. Both statements are therefore incorrect.

Hence, the correct answer is option 3) Both Statement I and Statement II are incorrect.

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NEET 2025

Ques: What is the pattern of inheritance for a polygenic trait?

Option 1) X-linked recessive inheritance pattern

Option 2) Mendelian inheritance pattern

Option 3) Non-Mendelian inheritance pattern

Option 4) Autosomal dominant pattern

Difficulty level - Difficult

Solution - Polygenic traits are affected by several genes, which are known as polygenes, with every gene having a small effect on the resultant expression of the trait, and that leads to a continuous variation in characters. Examples of such human traits are height, skin colour, and eye colour. As the pattern of inheritance is due to more than one gene and does not follow the simple dominant-recessive relationship of Mendelian characteristics, it is known as non-Mendelian inheritance. Environmental factors may also change the expression of these characteristics.

Hence, the correct answer is option 3) Non-Mendelian Inheritance Pattern.

Ques: Genes R and Y follow independent assortment. If RRYY produces round yellow seeds and rryy produces wrinkled green seeds, what will be the phenotypic ratio of the F_2 generation?

Option 1) Phenotypic ratio 9:7

Option 2) Phenotypic ratio 1:2:1

Option 3) Phenotypic ratio 3:1

Option 4) Phenotypic ratio 9:3:3:1

Difficulty level- Medium

Solution - Given -

- Parent 1: RRYY
- Parent 2: rryy

The F_1 generation will all be heterozygous RrYy and will display the dominant traits - round yellow seeds. Now,

- Parent 1: RrYy
- Parent 2: RrYy

The gametes for F_2 from each parent will be: RY, Ry, rY, ry.

After fertilisation, the phenotypes:

1. Round Yellow (RRYY, RRYy, RrYY, RrYy): 9
2. Round Green (RRyy, Rryy): 3
3. Wrinkled Yellow (rrYY, rrYy): 3
4. Wrinkled Green (rryy): 1

Hence, the correct answer is option 4) Phenotypic ratio 9:3:3:1.

NEET 2024

Ques: Match List I with List II

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List - I	List -II
A. Frederick Griffth	**I. Genetic code**
B. Francois Jacob & Jacque Monod	**II. Semi-conservative mode of DNA replication**
C. Har Gobind Khorana	**III. Transformation**
D. Meselson & Stahl	**IV.Lac operon**

Choose the correct answer from the options given below:

Option 1) A-III, B-II, C-I, D-IV

Option 2) A-III, B-IV, C-I, D-II

Option 3) A-II, B-III, C-IV, D-I

Option 4) A-IV, B-I, C-II, D-III

Difficulty - Medium

Operon:

- An operon is a unit of bacterial gene expression and regulation, including structural genes and control elements in DNA recognised by regulator gene product(s).
- In an operon, a polycistronic structural gene is regulated by a common promoter and regulatory genes.
- The elucidation of the lac operon was also a result of a close association between a geneticist, Francois Jacob, and a biochemist, Jacque Monod.

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Solution - A. Frederick Griffith → III. Transformation: Griffith demonstrated that genetic material could be passed from one strain of Streptococcus pneumoniae to another by discovering the phenomena of transformation in bacteria.

B. Jacques Monod & Francois Jacob → IV. Lac operon: Jacob and Monod introduced the idea of the Lac operon, which describes how E. coli's gene expression is controlled.

C. Har Gobind Khorana → I. Genetic code: Khorana made a substantial contribution to the deciphering of the genetic code, which determines how amino acids are specified by nucleotide sequences.

D. Meselson and Stahl → II. Semi-conservative mode of DNA replication: Meselson and Stahl experimentally supported the semi-conservative mode of DNA replication using isotope labelling.

Hence, the correct answer is option 2)A-III, B-IV, C-I, D-II.

Ques: Which one of the following can be explained based on Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles does not show any expression and both the characters appear as such in F_2 generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called a factor.

E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

Option 1) A, B and C only

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Option 2) A, C, D and E only

Option 3) B, C and D only

Option 4) A, B, C, D and E

Difficulty - Medium

Solution - According to the Law of Dominance

(1) Characters are controlled by discrete units called factors
(2) Factors occur in pairs
(3) In a dissimilar pair of factors one member of the pair dominates (dominant) the other recessive

The law of dominance is used to explain the expression of only one of the parental characters in a monohybrid cross.

The law of segregation is based on the fact that the alleles do not show any expression and both the characters are recovered as such in the F_2 generation.

Hence, the correct answer is option 2) A, C, D and E only

Ques: In a plant, black seed colour (BB/Bb) is dominant over white seed colour (bb). To find out the genotype of the black seed plant, which of the following genotypes will you cross?

Option 1) BB

Option 2) bb

Option 3) Bb

Option 4) BB/Bb

Difficulty - Medium

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Solution - In a test cross, an organism with an unknown genotype is mated to an organism with a known genotype to ascertain the genotype of the unknown organism. The test cross is said to have been created by Gregor Mendel. To determine the genotype of a black seed colour at F2, the black seed from F2 is crossed with the white seed colour. This is called a test cross. To determine the genotype of (BB/Bb) black seed we need to cross them with white seed (bb)

Hence, the correct answer is Option (2) bb.

Ques: Match List I with List II

List I	List II
A. Two or more alternative forms of a gene	**I. Back cross**
B. Cross of F1 progeny with homozygous recessive parent	**II.Ploidy**
C.Cross of F1 progeny with any of the parents	**III.Allele**
D. Number of chromosome sets in plant	**IV.Test cross**

Choose the correct answer from the options given below:

Option 1) A-I, B-II, C-III, D-IV

Option 2) A-II, B-I, C-III, D-IV

Option 3) A-III, B-IV, C-I, D-II

Option 4) A-IV, B-III, C-II, D-I

Difficulty - Medium

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Solution - **A. Two or more alternative forms of a gene → III. Allele**: Different variations of a gene located at the same location on a chromosome are referred to as alleles.

- **B. Cross of F1 progeny with homozygous recessive parent → IV. Test cross**: By mating the F1 hybrid with a homozygous recessive parent, a test cross is created to ascertain the genotype of the hybrid.
- **C. Cross of F1 progeny with any of the parents → I. Back cross**: A back cross involves crossing the F1 progeny with either of its parents.
- **D. Number of chromosome sets in a plant → II. Ploidy**: Ploidy refers to the number of complete sets of chromosomes in an organism.

Hence, the correct answer is option 3)A-III, B-IV, C-I, D-II.

NEET 2023

Ques: - The phenomenon of pleiotropism refers to

Option 1 - The presence of several alleles of a single gene controlling a single crossover.

Option 2 - Presence of two alleles, each of the two genes controlling a single trait.

Option 3 - A single gene affecting multiple phenotypic expressions.

Option 4 - more than two genes affecting a single character.

Difficulty: Medium

Solution: Pleiotropy is a genetic principle in which a solitary gene affects various, seemingly unrelated phenotypic traits. This occurs because the gene's product, typically a protein or enzyme, can influence numerous biological processes or tissues.

Instances of Pleiotropy:

1. Sickle Cell Anemia: This condition is attributed to a change in the HBB gene, which encodes haemoglobin. The altered gene leads to a distinctive red blood cell shape and confers resistance to malaria. Still, it also results in a range of health issues, including anaemia and potential organ damage.
2. Marfan Syndrome: Here, a mutation in the FBN1 gene influences connective tissue, causing several outcomes such as elongated limbs and fingers, heart issues and ocular concerns like lens displacement.
3. Phenylketonuria (PKU): A defect in the PAH gene alters the metabolism of phenylalanine, leading to a trio of consequences: intellectual impairment, alterations in skin pigmentation, and behavioural challenges.

Hence, the correct answer is option 3) a single gene affecting multiple phenotypic expressions.

Ques: - The frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes to map their position on a chromosome was used for the first time by

Option 1 - Thomas Hunt Morgan

Option 2 - Sutton and Boveri

Option 3 - Alfred Sturtevant

Option 4 - Henking

Difficulty: Medium

Solution: Alfred Sturtevant was a geneticist who played a significant role in genetics, particularly in mapping genes on chromosomes. He was the first to propose the use of recombination frequency as a measure of the distance between genes on the same chromosome. Sturtevant's work was based on the observations made by Thomas Hunt Morgan, who studied the patterns of inheritance in fruit flies. However, it was Sturtevant who developed the concept of genetic mapping using recombination

frequencies, which later became known as Sturtevant's mapping function. His groundbreaking work paved the way for the understanding of gene linkage and the construction of genetic maps.

Hence the correct answer is option 3) Alfred Sturtevant

Ques: - Broad plam with single palm crease is visible in a person suffering from –

Option 1 - Down's syndrome

Option 2 - Turner's syndrome

Option 3 - Klinefelter's syndrome

Option 4 - Thalassemia

Difficulty: Medium

Solution: A broad palm with a single palm crease, also known as a simian crease or a single transverse palmar crease, is commonly seen in individuals with Down's syndrome. Down's syndrome is a genetic disorder caused by the presence of an extra copy of chromosome 21. This condition is characterized by various physical and intellectual disabilities, and certain physical features are often associated with it.

One of the common physical features observed in individuals with Down's syndrome is a single transverse palmar crease, where the normal two creases in the palm fuse into one. This can give the appearance of a broad palm with a single crease extending across the palm.

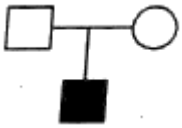
Hence, the correct answer is Option (1) Down's syndrome.

Ques: Which one of the following symbols represents mating between relatives in human pedigree analysis?

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Option 1 - 

Option 2 - 

Option 3 - 

Option 4 -  -

Difficulty: Medium

Solution: In human pedigree analysis, mating between relatives is represented by a double horizontal line connecting a male and a female. This symbol indicates consanguineous mating, which means that the individuals involved share a common ancestor. The double horizontal line represents a marriage or mating relationship, while the vertical lines extending downward from the double line represent their offspring.

Hence the correct answer is option 2.

Ques: Which of the following statements are correct about Klinefelter's Syndrome?

A. This disorder was first described by Langdon Down (1866).

B. Such an individual has overall masculine development. However, the feminine development is also expressed.

C. The affected individual is short-statured.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below.

Option 1 - A and B only

Option 2 - C and D only

Option 3 - B and E only

Option 4 - A and E only

Difficulty: Medium

Solution: Klinefelter Syndrome (KS) is a genetic disorder that affects males and is caused by the presence of an extra X chromosome. It occurs due to a random error during the formation of sperm or egg cells, leading to an extra X chromosome in the affected individual.

The correct statements about Klinefelter's Syndrome are:

B. Such an individual has overall masculine development. However, the feminine development is also expressed.
E. Such individuals are sterile.

Hence, the correct answer is option 3) B and E only

NEET 2022

Ques: Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason(R).
Assertion (A): Mendel's law of Independent assortment does not hold good for the genes located closely on the same chromosome.
Reason (R): Closely located genes assort independently.

In the light of the above statements, choose the correct answer from the options given below:

Option 1 - Both (A) and (R) are correct but (R) is not the correct explanation of (A)

Option 2 - (A) is correct but (R) is not correct

Option 3 - (A) is not correct but (R) is correct

Option 4 - Both (A) and (R) are correct and (R) is the correct explanation of (A)

Difficulty: Medium

Solution: The evaluation of the statements is as follows:
Statement 1: "If genes are closely located on the chromosome, they won't assort independently and do not show recombination."
This is correct. Those genes that lie near each other along the same chromosome are typically co-inherited since they may form a part of a more limited crossover region owing to genetic linkage.
Statement 2: "Mendel's law of independent assortment holds good only for the genes that are located on the same chromosome, not for the closely related genes."
This is false. Mendel's law of independent assortment holds for genes that are located on different chromosomes or far enough on the same chromosome. The closer the linked genes, the more dependent the genes' distribution will be. This cannot occur according to the law.
In summary, Statement 1 is correct, while Statement 2 is wrong.

Hence the correct answer is Option 2. (A) is correct but (R) is not correct

Ques: Which of the following occurs due to the presence of an autosome-linked dominant trait?

Option 1 - Myotonic dystrophy

Option 2 - Haemophilia

Option 3 - Thalassemia

Option 4 - Sickle cell anaemia

Difficulty: Medium

Solution: Hemophilia is a disorder in which blood doesn't clot normally and it is an X-linked recessive disorder.

Thalassemia is a blood disorder involving a lower-than-normal amount of an oxygen-carrying protein. It is an autosomal recessive disorder.

Sickle cell anemia is a group of disorders that causes red blood cells to become misshapen and break down it is an autosomal recessive disorder.

Myotonic dystrophy is an autosomal dominant disorder i.e.it occurs due to the presence of autosomal linked dominant trail.

Hence, the correct answer is option 1) Myotonic dystrophy.

Ques: Given below are two statements:

Statement I: Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance

Statement II: Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height

In the light of the above statements, choose the correct answer from the options given below:

Option 1 - Both Statement I and Statement II are incorrect.

Option 2 - Statement I is correct but Statement II is incorrect.

Option 3 - Statement I is incorrect but Statement II is correct.

Option 4 - Both Statement I and Statement II are correct.

Difficulty: Easy

Solution: George J. Mendel conducted hybridization experiments on garden peas and selected 14 true-breeding pea plant varieties (seven contrasting traits). Contrasting traits

Studied were: - 1.) Smooth or wrinkled seeds

2.) Yellow or green seeds

3.) Inflated or constricted pods

4.) Green or yellow pods.

5.) Tall or dwarf plants

6) Violet or white flowers

7.) Axial or terminal flower positions.

Hence, the correct answer is option 4)Both Statement I and Statement II are correct.

Ques: The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome?

Option 1 - d,b, a,c

Option 2 - a,b,c,d

Option 3 - a,c,b,d

Option 4 - a,d,b,c

Difficulty: Medium

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Solution: Distance between genes is measured by frequency of recombination.

According to the question,

-Recombination frequency between a & d is highest. This means they are farthest.

-So, options 2 & 3 can be our answer

a_____b

-Recombination frequency between

a and b = 20%

a and c = 5%

so, a and c are close.

Hence the sequence is

a c b d

Hence, Option (3) is correct. a,c,b,d

Ques: - XO type of sex determination can be found in:

Option 1 - Birds

Option 2 - Grasshoppers

Option 3 - Monkeys

Option 4 - Drosophila

Difficulty: Medium

Solution : Grasshopper - XO, XX Males females

X XX

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Birds - Males females

ZZ ZW

Drosophila - Male females

XY XX

Monkey - Male females

XY XX

Hence the correct option is (2). Grasshoppers

Ques: If a colour-blind female marries a man whose mother was also colour-blind, what are the chances of her progeny having colour-blindness?

Option 1 - 50%

Option 2 - 75%

Option 3 - 100%

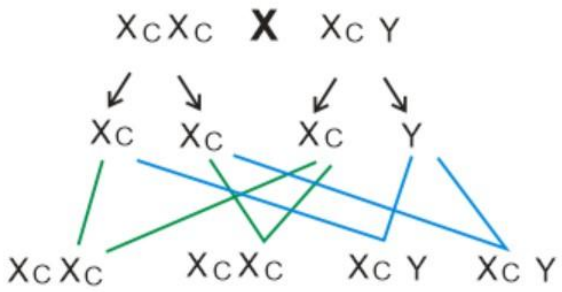
Option 4 - 25%

Difficulty: Medium

Solution: Colour blindness is the inability to perceive colours normally due to malfunctioning or missing cones (nerve cells) in the eyes. You could find it difficult to perceive the brightness of colours or to distinguish between different hues or tones. As given, the Female is colourblind i.e., XcXc & Male's mother was colourblind which means the male is XcY.

So,

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So, all the boys and girls will be colourblind. Hence 100% is the answer

Hence, the correct answer is Option (3) 100%

NEET 2021

Ques: - Experimental verification of the chromosomal theory of inheritance was done by:

Option 1 - Mendel

Option 2 - Sutton

Option 3 - Boveri

Option 4 - Morgan

Difficulty: Medium

Solution: The Chromosomal Theory of Inheritance, crucial in biology, establishes chromosomes as the carriers of genetic information responsible for Mendelian patterns of inheritance. This theory was founded on the works of Theodor Boveri and Walter Sutton. It aligns well with Mendel's laws and is bolstered by key observations:

1. Meiosis involves homologous chromosome pairs moving independently as distinct entities.
2. Parents create gametes with half their chromosome count.
3. Despite differences in size and shape, sperm and egg chromosomes are equal in

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number, indicating equal genetic input.
4. Upon fertilization, gametic chromosomes unite to form offspring with a chromosome count matching their parents.

Hence the correct answer is option 4) Morgan.

Ques: How many true-breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits

Option 1 - 4

Option 2 - 2

Option 3 - 14

Option 4 - 8

Difficulty: Medium

Solution: Seven pairs of contrasting features were represented by the 14 true-breeding pea plant varieties that Mendel chose. Having the exception of one character having two opposing forms (traits), each pair was identical in every way. Mendel's principles of inheritance were developed as a result of the study of inheritance patterns using these opposing features, such as tall vs. dwarf or round vs. wrinkled seeds.

Hence, the correct option is (c).

NEET 2020

Ques: In a cross between a male and female, both heterozygous for the sickle cell anaemia gene, what percentage of the progeny will be diseased?

Option 1 - 50%

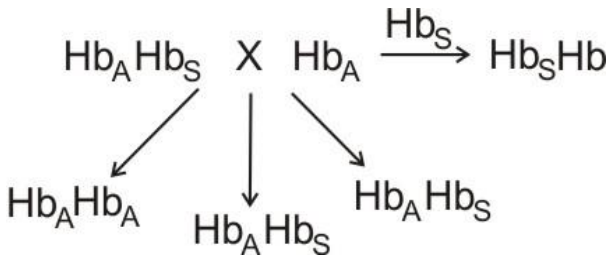
Option 2 - 75%

Option 3 - 25%

Option 4 - 100%

Difficulty: Medium

Solution :



Autosomal recessive inheritance is the mode of inheritance for sickle cell disease. Thus, an individual has to inherit two copies of the mutated HBB gene-one from each parent-to be affected. This will affect about 25% of the offspring when both parents are carriers (sickle cell trait). Carriers have genotype AS and are usually asymptomatic but can pass the gene to their children.

Hence the correct answer is Option 3) 25%

Ques: The production of gametes by the parents, the formation of zygotes, and the F_1 and F_2 plants, can be understood from a diagram called:

Option 1 - Bullet square

Option 2 - Punch square

Option 3 - Punnett square

Option 4 - Net square

Difficulty - Medium

Solution: The **Punnett square** is a diagrammatic tool used to predict the **genotypes** of offspring resulting from a particular genetic cross or breeding experiment. It was developed by **Reginald C. Punnett** in **1905**. The square helps biologists determine the

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probability of an offspring inheriting specific genotypes by organizing the potential genetic contributions from both parents. Each parent's alleles are represented along the top and side of the square, and the possible combinations of alleles are filled in the grid to show all potential genotypes for the offspring. This tool is especially useful in Mendelian genetics to understand inheritance patterns.

Hence, the correct answer is option 3, Punnett square

NEET 2019

Ques: A cell at the telophase stage is observed by a student in a plant brought from the field. He tells his teacher that this cell is not like other cells at the telophase stage. There is no form formation plate and thus the cell contains more nucleosomes as compared to other dividing cells. This would result in:

Option 1 - Aneuploidy

Option 2 - Polyploidy

Option 3 - Somaclonal variation

Option 4 - Polyteny

Difficulty: Medium

Solution: Euploidy -

Type of chromosomal numerical mutation due to gain or loss of one or more genomes.

2 types:-

1. Hypoploidy
2. polyploidy

When a cell contains slightly fewer chromosomes than a multiple of the monoploid number, it is said to be hypoploid, and when it has more than two sets of chromosomes, it is said to be polyploid:

Hence, the correct answer is Option (2) Polyploidy

Molecular Basis of Inheritance

The chapter "Molecular Basis of Inheritance" explains how genetic information is stored, replicated, and expressed in living organisms. It covers the (Watson and Crick model), and the (DNA → RNA → Protein). The important topics are discussed below:

Topic	Key Points
Structure of DNA	Watson and Crick's model of DNA
DNA replication	Process of DNA duplication
Central dogma	Flow genetic information: DNA → RNA → Protein

Transcription	Process of RNA synthesis from DNA
Translation	Protein synthesis from mRNA
Genetic code	Codon-based instructions for protein synthesis
Gene regulation	Control of gene expression in prokaryotes and eukaryotes
Operon Concept	Lac operon mechanism in prokaryotic gene regulation
Mutations	Changes in DNA sequence affecting gene function
Human genome project	Mapping and sequencing of human DNA
RNA Types	mRNA, tRNA, rRNA and their functions
DNA fingerprinting	Identification of individuals based on DNA patterns
Recombinant DNA technology	Genetic engineering techniques for modifying DNA

NEET 2026

Ques: In the lac operon, the z gene codes for:

Option 1) the repressor of lac operon

Option 2) transacetylase

Option 3) permease

Option 4) beta-galactosidase

Solution - The lacZ gene encodes beta-galactosidase, which hydrolyses lactose into glucose and galactose, enabling bacteria to use lactose as a carbon source. Together with lacY (permease) and lacA (transacetylase), lacZ forms the structural gene set of the lac operon.

Hence, the correct answer is option 4) beta-galactosidase.

Ques: Which of the following statements are correct with reference to a transcription unit?
A. A transcription unit in DNA is defined primarily by three regions: promoter, structural gene and terminator.
B. The promoter is said to be located towards the 5'-end of the structural gene.
C. The promoter is a DNA sequence that provides a binding site for RNA polymerase.
D. The promoter defines the template and coding strands.

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E. The terminator is located towards the 3′-end of the coding strand and it defines the end of the process of transcription.
Choose the correct answer from the options given below:

Option 1) B, C, D and E only

Option 2) A, B, C, D and E

Option 3) A, B, C and D only

Option 4) A, C, D and E only

Solution - A transcription unit has a promoter, a structural gene and a terminator. The promoter lies upstream at the 5′ end and binds RNA polymerase; it also determines the template and coding strands, while the terminator marks the end of transcription. All five statements are correct.

Hence, the correct answer is option 2) A, B, C, D and E.

Ques: Which of the following statements are correct with reference to the packaging of the DNA helix?
A. Histones are organised to form a unit of eight molecules, called a histone octamer.
B. Histones are negatively charged basic proteins.
C. Histones are rich in the basic amino acid residues - lysine and arginine.
D. The positively charged DNA is wrapped around the histone octamer to form a nucleosome.
E. The packaging of chromatin at higher levels requires an additional set of proteins called non-histone chromosomal proteins.
Choose the correct answer from the options given below:

Option 1) B, D and E only

Option 2) A, B and D only

Option 3) C, D and E only

Option 4) A, C and E only

Solution - Histones are positively charged proteins (not negatively charged) due to their lysine and arginine residues, which allow them to bind tightly to the negatively charged DNA. They form an octamer core around which DNA (which is negatively charged, not positive) wraps to form a nucleosome, and higher-order chromatin packaging requires additional non-histone chromosomal proteins.

Hence, the correct answer is option 4) A, C and E only.

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Re-NEET 2026

Ques: Which of the following statements about lac-operon is correct?

Option 1) Galactose can act as an inducer of lac operon

Option 2) Gene i is constitutively expressed

Option 3) Lactose activates repressor to bind to the operator

Option 4) Genes i, z, y and a share single common promoter

Solution - In the lac operon, gene i codes for the repressor protein and is constitutively (always) expressed, independent of the presence of an inducer.

Hence, the correct answer is option 2) Gene i is constitutively expressed.

Ques: Which of the following enzymes synthesizes precursor mRNA?

Option 1) DNA polymerase

Option 2) RNA polymerase I

Option 3) RNA polymerase II

Option 4) RNA polymerase III

Solution - In eukaryotes, RNA polymerase II synthesises precursor mRNA (hnRNA), which is later processed into mature mRNA. RNA polymerase I synthesises rRNA, while RNA polymerase III synthesises tRNA and 5S rRNA.

Hence, the correct answer is option 3) RNA polymerase II.

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NEET 2025

Ques: Which chromosome in the human genome has the highest number of genes?

Option 1) Chromosome 10

Option 2) Chromosome X
Option 3) Chromosome Y
Option 4) Chromosome 1
Difficulty Level: Easy
Solution: Chromosome 1 is the largest human chromosome and carries the most genes of all the chromosomes. It has more than 2,000 protein-coding genes that play a wide variety of fundamental biological functions. Chromosome X also carries many genes, particularly for sex-linked characteristics, but not as many as chromosome 1. Chromosome 10 has fewer genes than chromosome 1, and "Chromosome Y" has fewer than 100 genes. Thus, chromosome 1 has the most genes in the human genome.

Hence, the correct answer is option 4) Chromosome 1.

Ques: Match List I with List II:

List I	**List II**
A. Alfred Hershey and Martha Chase	**I. Streptococcus pneumoniae**
B. Euchromatin	**II. Densely packed and dark-stained**
C. Frederick Griffith	**III. Loosely packed and light-stained**

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D. Heterochromatin	**IV. DNA as genetic material confirmation**

Option 1: A-III, B-II, C-IV, D-I

Option 2: A-II, B-IV, C-I, D-III

Option 3: A-IV, B-II, C-I, D-III

Option 4: A-IV, B-III, C-I, D-II

Ques: Which of the following are the posttranscriptional events in an eukaryotic cell?
A. Transport of pre-mRNA to the cytoplasm prior to splicing.
B. Removal of introns and joining of exons.
C. Addition of methyl group at the 5′ end of hnRNA.
D. Addition of adenine residues at the 3′ end of hnRNA.
E. Base pairing of two complementary RNAs.
Difficulty Level: Medium
Option 1) C, D, E only
Option 2) A, B, C only
Option 3) B, C, D only
Option 4) B, C, E only
Solution: In eukaryotic cells, the first RNA transcript (hnRNA) goes through various post-transcriptional modifications before becoming functional mRNA. These include:

- Removal of introns and splicing of exons (B)
- Addition of a methylated cap at the 5′ end (C)
- Addition of a poly-A tail (adenine residues) at the 3′ end (D)

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Statement A is not correct since splicing takes place in the nucleus before the transport of the RNA to the cytoplasm.
Statement E is not relevant to normal mRNA processing, as it refers to RNA.

Hence, the correct answer is option 3) B, C, D only

Ques: Given below are two statements:

Statement I: Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II: RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

Difficulty level: Difficult

Option 1) Statement I is incorrect, but Statement II is correct.
Option 2) Both Statement I and Statement II are correct.
Option 3) Both Statement I and Statement II are incorrect.
Option 4) Statement I is correct, but Statement II is incorrect**.**
Solution: Statement I is false because both rRNA and tRNA interact with mRNA during the process of translation. The ribosome is present where the mRNA is attached, and it has a component that is made of rRNA. At this point, tRNA molecules deliver the correct amino acids by matching their anticodons with the codons on the mRNA.

Statement II is accurate because RNA interference (RNAi) is an important cellular defence mechanism that is present in all eukaryotic life. It consists of small double-stranded RNA molecules, like siRNA or miRNA, which block gene expression by binding to and neutralising the targeted mRNA molecules.

Hence, the correct answer is Option 1) Statement I is incorrect, but Statement II is correct.

Ques: Which factor is important for the termination of transcription?

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Difficulty: Medium

Option 1) γ (gamma)

Option 2) α (alpha)

Option 3) σ (sigma)

Option 4) ρ (rho)

Solution: Transcription termination in prokaryotes often includes some specific factors. The ρ (rho) factor is a protein that binds to the RNA and moves along it until it catches up with the RNA polymerase, causing the enzyme to dissociate from the DNA and thus terminating transcription. Other factors like α (alpha), γ (gamma), and σ (sigma) have different roles in transcription initiation or enzyme structure, not in the process of termination.

Hence, the correct answer is option (4) ρ (rho).

Ques: Histones are enriched with-

Difficulty level: Medium

Option 1) Phenylalanine & Arginine

Option 2) Lysine & Arginine

Option 3) Leucine & Lysine

Option 4) Phenylalanine & Leucine

Solution: The histones contain a high percentage of basic amino acids, mainly **lysine and arginine**, which are positively charged. The positive charge allows histones to strongly interact with the negatively charged phosphate backbone of the DNA to make a stable DNA-protein complex. These histones then combine to give a nucleosome,

which is the fundamental building block of chromatin, in which the DNA is wrapped around a histone protein core.

Hence, the correct answer is option(2) Lysine and Arginine.

Ques: Who proposed that the genetic code for amino acids should be made up of three nucleotides?

Difficulty level: Medium

Option 1) Franklin Stahl

Option 2) George Gamow

Option 3) Francis Crick

Option 4) Jacque Monod

Solution: George Gamow initially proposed that the genetic code has to be made up of triplets, which means groups of three nucleotide bases, to represent an amino acid. This is because there are four bases (A, T, C, G) and 20 amino acids to encode for. A base alone ($4^1 = 4$) or even pairs ($4^2 = 16$) is not sufficient to account for all 20 amino acids. Yet, triplets ($4^3 = 64$ combinations) offer more than sufficient codons, allowing each amino acid to be coded.

Hence, the correct answer is option 2) George Gamow.

NEET 2024

Ques: 1 Match List I with List II :

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	List I		List II
A	RNA polymerase III	**I**	snRNPs
B	Termination of transcription	**II**	Promotor
C	Splicing of Exons	**III**	Rho factor
D	TATA box	**IV**	snRNAs, tRNA

Choose the correct answer from the options given below :

Option 1) A-II, B-IV, C-I, D-III

Option 2) A-III, B-II, C-IV, D-I

Option 3) A-III, B-IV, C-I, D-II

Option 4) A-IV, B-III, C-I, D-II

Difficulty - Easy

Solution - In eukaryotes, RNA polymerase III codes for snRNAs, tRNA, and 5s rRNA.

• Splicing of exons is performed by snRNPs.
• TATA box is present in the promoter region of the transcription unit.
• The Rho factor is responsible for the termination of transcription.

Hence the correct answer is Option 4. A-IV, B-III, C-I, D-II.

Ques:2 Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of:

Option 1) 8 bp

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Option 2) 6 bp

Option 3) 4 bp

Option 4) 10 bp

Difficulty - Medium

Solution - The first restriction endonuclease – Hind II, whose functioning depends on a specific DNA nucleotide sequence was isolated. It was found that Hind II always cut DNA molecules at a particular point by recognizing a sequence of six base pairs. Option (1), (3), and (4) are incorrect because they have either more than 6 or less than 6 bp.

Hence, the correct answer is option (2)6 bp.

NEET 2023

Quest 3: Which one of the following is sequence on the corresponding coding strand, if the sequence on mRNA formed is as follows

5'AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3'?

Option 1 - 5'UAGCUAGCUAGCUAGCUAGCUAGC UAGC 3'

Option 2 - 3'UAGCUAGCUAGCUAGCUAGCUAGCUAGC 5'

Option 3 - 5'ATCGATCGATCGATCGATCGATCGATCG 3'

Option 4 - 3'ATCGATCGATCGATCGATCGATCGATCG 5'

Difficulty: Easy

Solution: To find the corresponding coding strand, we replace each base according to the base pairing rules:

A (mRNA) pairs with U (coding DNA)
U (mRNA) pairs with A (coding DNA)

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C (mRNA) pairs with G (coding DNA)
G (mRNA) pairs with C (coding DNA)

Applying these base pairing rules to the given mRNA sequence, we get:

5′AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3′
3′TAGCTAGCTAGCTAGCTAGCTAGC TAGC 5′

Therefore, the correct corresponding coding strand sequence is 3′TAGCTAGCTAGCTAGCTAGCTAGC TAGC 5′.

Hence, the correct answer is option 3 - 5′ATCGATCGATCGATCGATCGATCG 3′

Quest: 4 - Given below are two statements:

Statements I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form a nucleosome.

In the light of the above statements, choose the **correct** answer from the options given below:

Option 1 - Both Statement I and Statement II are true.

Option 2 - Both Statement I and Statement II are false.

Option 3 - Statement I is correct but statement II is false.

Option 4 - Statement I is incorrect but statement II is true.

Difficulty: Difficult

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Solution: Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

This statement is incorrect. In prokaryotes, the DNA is not associated with proteins like histones to form nucleosomes as seen in eukaryotes. Instead, prokaryotic DNA is typically found in a region called the nucleoid, where it is condensed and organized to some extent, but without the association of histones or other proteins.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form a nucleosome.

This statement is true. In eukaryotes, DNA is wrapped around histone proteins to form nucleosomes. The DNA molecule is negatively charged due to its phosphate backbone, while the histone proteins are positively charged. The interaction between the negatively charged DNA and the positively charged histones allows for the compaction and organization of DNA into a higher-order structure.

Hence, the correct answer is Option 4) Statement I is incorrect, but statement II is true.

Ques: 5 - Expressed Sequence Tags (ESTs) refer to

Option 1 - All genes that are expressed as RNA.

Option 2 - All genes are expressed as proteins.

Option 3 - All genes whether expressed or unexpressed.

Option 4 - Certain important expressed genes.

Solution: ESTs are short DNA sequences from cDNA libraries produced, in this case, by a fragment of mRNA, representing the transcription of expressed genes in specific tissues or organisms. ESTs can be produced by first obtaining a sample of mRNA which then undergoes reverse transcription, an enzyme that helps generate cDNA, into a vector and then is sequenced. ESTs particularly play a significant role in discovering

genes and gene expression experiments where the identity and some kinds of expression patterns associated with genes can help scientists investigate the genetics and biological processes involved within any disease.

Hence, the correct answer is Option 1) All genes that are expressed as RNA.

Ques: 5 - Match **List I** with **List II**.

	List I		**List II**
A.	Gene 'a'	I.	β-galactosidase
B.	Gene 'y'	II.	Transacetylase
C.	Gene 'I'	III.	Permease
D.	Gene 'z'	IV.	Repressor protein

Choose the **correct** answer from the options given below:

Option 1 - A-II, B-I, C-IV, D-III

Option 2 - A-II, B-III, C-IV, D-I

Option 3 - A-III, B-IV, C-I, D-II

Option 4 - A-III, B-I, C-IV, D-II

Difficulty: Medium

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Solution: A. Gene 'a' - II. Transacetylase
Transacetylase is an enzyme involved in the metabolism of lactose.

B. Gene 'y' - III. Permease
Permease is a protein that facilitates the movement of molecules across the cell membrane.

C. Gene 'I' - IV. Repressor protein
Repressor protein regulates the expression of genes by binding to the operator region of DNA and preventing transcription.

D. Gene 'z' - I. β-galactosidase
β-galactosidase is an enzyme that hydrolyzes lactose into glucose and galactose.

Hence, the correct answer is Option 2 A-II, B-III, C-IV, D-I

Ques: 6 - What is the role of RNA polymerase III in the process of transcription in Eukaryotes?

Option 1 - Transcription of tRNAs (28S, 18S AND 5.8S)

Option 2 - Transcription of tRNA, 5 srRNA and snRNA

Option 3 - Transcription of the precursor of mRNA

Option 4 - Transcription of only snRNAs

Difficulty: Medium

Solution: The correct option is Option 2) Transcription of tRNA, 5S rRNA, and snRNA.

RNA polymerase III is an enzyme responsible for transcribing specific types of RNA molecules in eukaryotes. Its primary role is the transcription of three major types of RNA:

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tRNA (transfer RNA): tRNA molecules play a crucial role in protein synthesis by carrying specific amino acids to the ribosomes. RNA polymerase III transcribes the genes encoding tRNA molecules.

5S rRNA (5S ribosomal RNA): 5S rRNA is a component of the large ribosomal subunit involved in protein synthesis. RNA polymerase III transcribes the genes encoding 5S rRNA.

snRNA (small nuclear RNA): snRNA molecules are involved in the processing of pre-mRNA and in the assembly of spliceosomes, which are responsible for removing introns from pre-mRNA. RNA polymerase III transcribes the genes encoding snRNA.

Therefore, RNA polymerase III plays a critical role in transcribing tRNA, 5S rRNA, and snRNA, contributing to important cellular functions such as protein synthesis and RNA processing.

Ques: 7 - Unequivocal proof that DNA is the genetic material was first proposed by

Option 1 - Frederick Griffith

Option 2 - Alfred Hershey and Martha Chase

Option 3 - Avery, Macleod and McCarty

Option 4 - Wilkins and Franklin

Difficulty: Easy

Solution: Alfred Hershey and Martha Chase conducted the Hershey-Chase experiment in 1952, which provided unequivocal proof that DNA is the genetic material. In their experiment, they used bacteriophages, which are viruses that infect bacteria. They labelled the DNA of the bacteriophage with radioactive phosphorus-32 and the protein coat with radioactive sulfur-35.

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By infecting bacteria with these labelled bacteriophages and then separating the viral DNA and proteins from the infected bacteria, Hershey and Chase showed that only the radioactive DNA, not the protein, was transferred to the next generation of phages. This demonstrated that DNA, not protein, carried the genetic information and served as the hereditary material.

Their experiment provided strong evidence supporting the idea that DNA, rather than protein, is the genetic material responsible for transmitting inherited traits. Therefore, Hershey and Chase were the ones who first proposed the unequivocal proof that DNA is the genetic material.

Hence, the correct answer is Option 2) Alfred Hershey and Martha Chase.

NEET 2022

Ques: 8 - If a geneticist uses the blind approach for sequencing the whole genome of an organism, followed by assignment of function to different segments, the methodology adopted by him is called:

Option 1 - Gene mapping

Option 2 - Expressed sequence tags

Option 3 - Bioinformatics

Option 4 - Sequence annotation

Difficulty: Medium

Solution: Sequencing the whole set of genomes that contained all the coding and non-coding sequences and later assigning different regions in the sequence with functions is called sequence annotation.

The second approach is EST (Expressed Sequence Tags) in which the focus is on identifying all the genes that are expressed as RNA.

Hence, the correct option is (4) Sequence annotation

Ques: 9 - The process of translation of mRNA to proteins begins as soon as :

Option 1 - The larger subunit of ribosome encounters mRNA

Option 2 - Both the subunits join together to bind with mRNA

Option 3 - The tRNA is activated and the larger subunit of ribosome encounters mRNA

Option 4 - The small subunit of ribosome encounters mRNA

Difficulty: Medium

Solution: When the smaller submit of ribosome encounters m-RNA, the process of translation of m-RNA to protein begins. This process is followed by the binding of a larger subunit. The activated t-RNA molecule carrying the amine acid methionine binds to the start codon of the mRNA sequence.

Hence, the correct option is (4). The small subunit of ribosome encounters mRNA

Ques: 9 - Read the following statements and choose the set of correct statements :

(a) Euchromatin is loosely packed chromatin

(b) Heterochromatin is transcriptionally active

(c) Histone octamer is wrapped by negatively charged DNA in the nucleosome

(d) Histones are rich in lysine and arginine

(e) A typical nucleosome contains 400bp400bp of DNA helix

Choose the correct answer from the options given below:

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Option 1 - (a), (c), (d) Only

Option 2 - (b), (e) Only

Option 3 - (a), (c), (e) Only

Option 4 - (b), (d), (e) Only

Difficulty: Medium

Solution: **The correct statements are:**

- **(a)** Euchromatin is loosely packed chromatin.
- **(c)** Histone octamer is wrapped by negatively charged DNA in a nucleosome.
- **(d)** Histones are rich in lysine and arginine.

Explanation:

- **(b)** is incorrect because heterochromatin is generally transcriptionally inactive.
- **(e)** is incorrect because a typical nucleosome contains about 200 base pairs (bp) of DNA, not 400 bp.

Correct answer: (a), (c), and (d)

Hence, the correct answer is option 1) (a), (c), and (d).

Ques: 10 - If the length of a DNA molecule is 1.1 meters, what will be the approximate number of base pairs?

Option 1 - $6.3 \text{ X } 10^{-9}$bp

Option 2 - $3.6 \text{ X } 10^{-9}$bp

Option 3 - $6.6 \text{ X } 10^{6}$bp

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Option 4 - 3.3 X 10^{-9}bp

Difficulty: Medium

Solution: It is estimated that a DNA molecule, which is around 1.1 meters long, has nearly 3.23 billion base pairs. This amount is calculated using the roughly 0.34-nanometre length of a single base pair in DNA. By converting the DNA length into nanometers **1.1 meters(=1.1×10^{9})nanometers** and dividing it by the length of a single base pair, we can estimate the total number of base pairs.

Hence, the correct answer is option 4.

Ques: 11 - In an E. coli Strain i gene gets mutated and its product can not bind the inducer molecule. If a growth medium is provided with lactose, what be the outcome?

Option 1 - z,y, a gene will be transcribed

Option 2 - z,y, a gene will not be translated

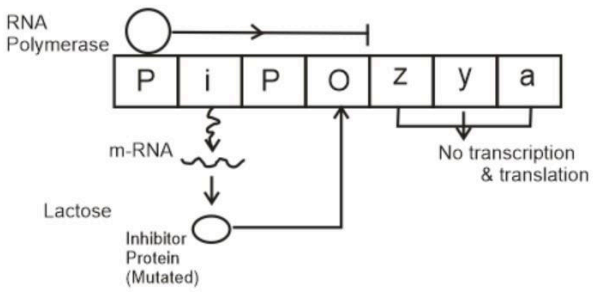
Option 3 - RNA polymerase will bind the promoter region

Option 4 - Only the z gene will get transcribed

Difficulty: Medium

Solution: Normally, in On condition i.e. when lactose is present in the medium, the inhibitor protein binds to the lactose and not to the operator gene. This is because the inhibitor protein has more affinity with lactose as compared to the operator gene. This will allow the z, and y & genes to undergo transcription & then translation.

In this case, the 'i' gene' has been mutated due to which, even in the presence of lactose, inhibitor protein binds to the operator gene which will block the movement of RNA polymerase, and z, y, and genes will not undergo transcription and translation.



Hence the correct option is 3. RNA polymerase will bind the promoter region

NEET 2021

Ques: 12 - Which one of the following statements about Histones is wrong?

Option 1 - Histones are organized to form a unit of 8 molecules.

Option 2 - The pH of histones is slightly acidic

Option 3 - Histones are rich in amino acids- Lysine and Arginine

Option 4 - Histones carry a positive charge in the side chain.

Difficulty: Medium

Solution: Histones are rich in basic amino acids like lysine and arginine which carry positive charge in their side chains due to which they are positively charged. This positive charge allows histones to interact with the negatively charged DNA which facilitates the formation of nucleosomes and helps in the packaging of DNA in the nucleus.

Hence, the correct answer is option 2) The pH of histones is slightly acidic

Ques: 13 - **Statement I:**

The codon 'AUG' codes for methionine and phenylalanine.

Statement II :

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'AAA' and 'AAG are both codons that code for the amino acid lysine.

In light of the above statements, choose the **correct** answer from the options given below.

Option 1 - Both Statement I and Statement II are true

Option 2 - Both Statement I and Statement II are false

Option 3 - Statement I is correct but Statement II is false

Option 4 - Statement I is incorrect but Statement II is true.

Difficult: Medium

Solution: AUG codes for only methionine but not tryptophan. (A given codon codes for only one amino acid but not more than one amino acid.). AAG and AAA are the two codons for the amino acid Lysine.

Hence, the correct option is 4 - Statement I is incorrect but Statement II is true.

Ques: 14 - If Adenine makes 30% of the DNA molecule, what will be the percentage of Thymine, Guanine, and Cytosine in it?

Option 1 - $T:20:G:30:C:20$

Option 2 - $T:20:G:20:C:30$

Option 3 - $T:30:G:20:C:20$

Option 4 - $T:20:G:25:C:25$

Difficulty: Medium

Solution: $A=T\ T=30\%$

$A+T=60\%$

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$G + C = 100 - 60 = 40$

$\therefore G = C = 20\%$

Hence, the correct option is (3) $T : 30 : G : 20 : C : 20$

Ques: 15 - Identify the correct statement:

Option 1 - In capping, methyl guanosine triphosphate is added to the 3′ end of hnRNA

Option 2 - RNA polymerase binds with Rho factor to terminate the process of transcription in bacteria.

Option 3 - The coding strand in a transcription unit is copied to an mRNA

Option 4 - Split gene arrangement is characteristic of prokaryotes.

Difficulty: Medium

Solution: In the cell, Rho binds to untranslated naked RNAs and terminates the synthesis of mRNA at the end of a significant number of operons.

Hence, the correct option is 2.

Ques: 16 - Which of the following RNAs is not required for the synthesis of protein?

Option 1 - mRNA

Option 2 - tRNA

Option 3 - rRNA

Option 4 - siRNA

Difficulty: Medium

Solution: siRNA (small interfering RNA) is a type of RNA involved in the regulation of gene expression, particularly in RNA interference (RNAi), where it silences specific

gene expression by degrading mRNA. mRNA (messenger RNA) is synthesized during transcription and carries genetic information from DNA to ribosomes for protein synthesis. tRNA (transfer RNA) assists in translation by bringing amino acids to the ribosome, aligning with the mRNA codons to build a protein. rRNA (ribosomal RNA) forms the core of ribosome structure and catalyzes protein synthesis by facilitating the binding of tRNA to mRNA during translation.

Hence, the correct answer is option 4 - siRNA

Ques: 17 - Which is the "only enzyme" that has the capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes?

Option 1 - DNA-dependent DNA polymerase

Option 2 - DNA-dependent RNA polymerase

Option 3 - DNA Ligase

Option 4 - DNase

Difficulty: Medium

Solution: In the process of **transcription** in **prokaryotes**, **DNA-dependent RNA polymerase** plays a crucial role in catalyzing all stages: **initiation**, **elongation**, and **termination**. The enzyme binds to the **DNA template** at the **promoter region**, where it begins **synthesizing an RNA molecule** that is complementary to the DNA strand. This results in the production of **RNA transcripts** that carry genetic information from the DNA to be used in protein synthesis. The RNA polymerase moves along the DNA, adding nucleotides to the growing RNA chain during elongation, and eventually releases the RNA molecule once it reaches a **termination signal**, completing the process.

Hence, the correct answer is option 2) DNA-dependent RNA polymerase

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Ques: 18 - What is the role of RNA polymerase III in the process of transcription in eukaryotes?

Option 1 - Transcribe rRNAs (28S, 18S and 5.8S)

Option 2 - Transcribes tRNA, 5S rRNAA and snRNA.

Option 3 - Transcribes precursor of mRNA

Option 4 - Transcribes only snRNAs

Difficulty: Difficult

Solution:

Eukaryotic cells need RNA polymerase III (Pol III) to transcribe DNA and synthesize essential small RNAs. These small RNAs include ribosomal 5S rRNA and transfer RNA. The transcripts produced by Pol III are essential for the cell's necessary activities, such as protein synthesis. The genes transcribed by Pol III are called "housekeeping" genes because they are constitutively expressed in all cell types and under different environmental conditions, thus preserving cellular function and homeostasis.

Hence, the correct answer is Option (2) Transcribes tRNA, 5S rRNA, and snRNA.

Ques: 19 - Which one of the following is the starter codon?

Option 1 - AUG

Option 2 - UGA

Option 3 - UAA

Option 4 - UAG

Difficulty: Medium

Solution: As we learnt in

Salient features of genetics code -

1. The codon is triplet, 61 codons code for amino acids and 3 codons do not code for any amino acids, hence they function as stop codons.

Hence, AUG is the start codon.

Ques: 20 - Mutations in plant cells can be induced by:

Option 1 - Kinetin

Option 2 - Infrared rays

Option 3 - Gamma rays

Option 4 - Zeatin

Difficulty: Medium

Solution:

Gamma rays are used in inducing mutations in seeds and other planting materials such as cuttings, pollen, or tissue culture. Gamma rays are a type of ionizing radiation commonly used to induce mutations in seeds, cuttings, pollen, or tissue cultures to create genetic diversity. By altering the DNA in these materials, gamma rays can result in beneficial traits such as improved disease resistance or increased yield. This method is widely used in plant breeding programs to develop new varieties of crops with desirable characteristics.

Hence, the correct answer is option 3) Gamma rays

Evolution

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The chapter "Evolution" explains the origin and development of life on Earth through gradual changes over time.

Topic	Key Points
Origin Of Life	Gradual development of life on Earth
Theories Of Evolution	Lamarck's Theory (Use And Disuse), Darwin's Theory Of Natural Selection(Natural Selection)
Natural Selection	Survival of the fittest, adaptation to the environment
Adaptive Radiation	Evolution of different species from a common ancestor
Evidence Of Evolution	Fossils, Embryology, Homologous And Analogous Organs
Hardy-Weinberg Principle	Genetic equilibrium in a population
Mechanisms Of Evolution	Genetic Drift, Mutation, Recombination, Natural Selection
Human Evolution	Evolutionary history of humans
Biodiversity And Adaptation	How species evolve and adapt to their environments

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NEET 2026

Ques: Since the origin and diversification of life on Earth, there have been five episodes of mass extinction of species. How is the sixth extinction, which is in progress, different from the previous episodes?

Option 1) The current species extinction rates are far lower than those in previous episodes.

Option 2) The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Option 3) The present net species extinction rate is zero.

Option 4) The current species extinction rate is nearly 10 times faster than that in previous episodes.

Solution - Earth has witnessed five major mass extinctions in the past, caused by natural events such as volcanic eruptions, asteroid impacts and climate shifts. The ongoing, sixth extinction is largely human-driven, with an estimated extinction rate 100-1000 times higher than the natural background rate, due to deforestation, habitat loss, pollution, overexploitation and climate change.

Hence, the correct answer is option 2) The present species extinction rates are 100 to 1000 times faster than in pre-human times.

Ques: Which of the following is not an example of convergent evolution?

Option 1) Fore limbs of whales and bats

Option 2) Flippers of penguins and dolphins

Option 3) Eyes of octopuses and mammals

Option 4) Wings of butterflies and birds

Solution - Forelimbs of whales and bats are homologous structures (an example of divergent evolution from a common ancestral pentadactyl limb), not convergent evolution.

Hence, the correct answer is option 1) Fore limbs of whales and bats.

Ques: Match List I with List II with respect to chronology of evolution of life forms:
List I - List II
A. About 65 mya - I. Jawless fish probably evolved
B. About 500 mya - II. The dinosaurs suddenly disappeared from the earth
C. About 350 mya - III. Seaweeds and few plants probably existed
D. About 320 mya - IV. Invertebrates were formed and became active
Choose the correct answer from the options given below:

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Option 1) A-III, B-IV, C-I, D-II

Option 2) A-II, B-IV, C-III, D-I

Option 3) A-II, B-IV, C-I, D-III

Option 4) A-I, B-II, C-III, D-IV

Solution - About 65 million years ago, the dinosaurs suddenly disappeared from the Earth. Around 500 million years ago, invertebrates were formed and became active. Jawless fish probably evolved around 350 million years ago, while seaweeds and a few plants probably existed around 320 million years ago.

Hence, the correct answer is option 3) A-II, B-IV, C-I, D-III.

Ques: Evolution of human appears parallel to the progressive development of brain and language skills. As such, the evolution of individual species in the sequence of their appearance is:

Option 1) Ramapithecus to Homo habilis to Homo erectus to Neanderthal to Homo sapiens

Option 2) Neanderthal to Ramapithecus to Homo habilis to Homo erectus to Homo sapiens

Option 3) Homo habilis to Homo erectus to Ramapithecus to Neanderthal to Homo sapiens

Option 4) Homo sapiens to Ramapithecus to Homo habilis to Neanderthal to Homo erectus

Solution - The correct evolutionary sequence is: Ramapithecus (earliest ape-man ancestor) to Homo habilis (first tool maker) to Homo erectus (upright man) to Neanderthal man (more advanced) to Homo sapiens (modern humans).

Hence, the correct answer is option 1) Ramapithecus to Homo habilis to Homo erectus to Neanderthal to Homo sapiens.

Re-NEET 2026

Ques: Adaptive radiation in placental mammals and Australian Marsupials leading to similarity between distant species is an example of -------.

Option 1) genetic drift

Option 2) divergent evolution

Option 3) convergent evolution

Option 4) founder effect

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Solution - Placental mammals and Australian marsupials evolved separately but developed similar traits due to similar environmental pressures, which is convergent evolution - unrelated species independently developing similar characteristics. A classic example is the placental wolf and the Tasmanian wolf (marsupial), which look similar despite being unrelated.

Hence, the correct answer is option 3) Convergent evolution.

Ques: Given below are two statements:
Statement I: Modern Homo sapiens arose in Australia and moved across continents.
Statement II: Homo sapiens arose around 75000 to 10000 years ago.
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1) Statement I is incorrect but Statement II is correct

Option 2) Both Statement I and Statement II are correct

Option 3) Both Statement I and Statement II are incorrect

Option 4) Statement I is correct but Statement II is incorrect

Solution - Modern Homo sapiens arose in Africa, not Australia, and moved across continents from there. Homo sapiens is estimated to have arisen around 75,000 to 10,000 years ago.

Hence, the correct answer is option 1) Statement I is incorrect but Statement II is correct.

Ques: Which of the following is not evidence for evolution?

Option 1) Divergent evolution of anatomical structures such as forelimbs

Option 2) Convergent evolution of traits like wings of birds and butterflies

Option 3) Paleontological evidence from fossil records

Option 4) Embryological support for evolution as proposed by Ernst Heckel

Solution - Embryological evidence based on Haeckel's biogenetic law ('ontogeny recapitulates phylogeny') is now considered largely inaccurate and is not reliable evidence for evolution. Divergent evolution, convergent evolution and paleontological fossil records are all accepted as genuine evidence for evolution.

Hence, the correct answer is option 4) Embryological support for evolution as proposed by Ernst Heckel.

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Ques: A population of diploid organisms is at Hardy-Weinberg equilibrium. If the frequency of allele A is 0.1, the frequency of AA is -----.

Option 1) 0.99

Option 2) 0.01

Option 3) 0.02

Option 4) 0.10

Solution - By the Hardy-Weinberg law, the frequency of the AA genotype is p^2. Given the frequency of allele A (p) is 0.1, frequency of AA = (0.1)^2 = 0.01.

Hence, the correct answer is option 2) 0.01.

Ques: Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Forelimbs of human and bats are homologous.
Reason R: Forelimbs of humans and bats have similar anatomical structure.
In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1) A is false but R is true

Option 2) Both A and R are correct and R is the correct explanation of A

Option 3) Both A and R are true, but R is not the correct explanation of A

Option 4) A is true but R is false

Solution - Forelimbs of humans and bats are indeed homologous structures - they share the same basic skeletal framework (humerus, radius, ulna, carpals) despite performing different functions. This shared anatomical structure is precisely what makes them homologous, so R correctly explains A.

Hence, the correct answer is option 2) Both A and R are correct and R is the correct explanation of A.

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NEET 2025

Ques: Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution.

Difficulty level: Difficult
Option 1) Analogy, divergent
Option 2) Analogy, convergent
Option 3) Homology, divergent
Option 4) Homology, convergent

Solution: Sweet potato and potato both perform the same function of food storage, but they belong to two different plant parts. Sweet potato originates from the adventitious roots, whereas potato arises from underground stems. This makes them an example of analogous organs that perform the same function but with a different structure and evolutionary origin. These types of organs develop because of convergent evolution, where organisms perform the same function but their origin are different.

Hence, the correct answer is Option (2) Analogy, convergent.

NEET 2024

Ques - 1 Match List I with List II :

	List I		List II
A	Mesozoic Era	**I**	Lower invertebrates
B	Proterozoic Era	**II**	Fish & Amphibia
C	Cenozoic Era	**III**	Birds & Reptiles

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D	Palaeozoic Era	**IV**	Mammals

Choose the correct answer from the options given below :

Option 1 - A-II, B-I, C-III, D-IV

Option 2 - A-III, B-I, C-II, D-IV

Option 3 - A-I, B-II, C-IV, D-III

Option 4 - A-III, B-I, C-IV, D-II

Difficulty: Medium

Solution: The Mesozoic Era, sometimes referred to as the "Age of Reptiles," is distinguished by the predominance of birds and reptiles.
The right match is A → III.

The Proterozoic Era is when early life, particularly lower invertebrates, first appeared.
B → I is the correct match.

Known as the "Age of Mammals," the Cenozoic Era saw mammals take over as the predominant form of life.
C → IV is the correct match.

Fish and amphibia rose to prominence throughout the Paleozoic Era.
D → II is the correct match.
Hence, the correct answer is option 4)A-III, B-I, C-IV, D-II.

Ques - 2 Given below are some stages of human evolution.
Arrange them in the correct sequence. (Past to Recent)

A. Homo habilis
B. Homo sapiens

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C. Homo neanderthalensis
D. Homo erectus

Choose the correct sequence of human evolution from the options given below:

Option 1 - D-A-C-B

Option 2 - B-A-D-C

Option 3 - C-B-D-A

Option 4 - A-D-C-B

Difficulty: Medium

Solution: The right sequence of human evolution is Homo habilis, then Homo erectus, then Homo neanderthalensis, and finally Homo sapiens. Homo habilis was the first human who used tools, while Homo erectus demonstrated a more efficient gait and a larger brain. Neanderthals were contemporary with early modern humans, and they represent the last stage of this evolutionary process, characterized by the highest level of cognitive ability.

Hence, the correct answer is Option 4)A-D-C-B

Ques - 3 The flippers of the Penguins and Dolphins are the example of the

Option 1 - Adaptive radiation

Option 2 - Natural selection

Option 3 - Convergent evolution

Option 4 - Divergent evolution

Difficulty: Medium

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Solution: The flippers of **penguins** and **dolphins** are an example of **convergent evolution**. In this process, unrelated species develop similar traits or adaptations due to having to adapt to similar environmental challenges or ecological niches. Although penguins (birds) and dolphins (mammals) are from different evolutionary lineages, both have evolved **flippers** as specialized limbs for efficient swimming in water. This is a result of **natural selection** favouring traits that enhance swimming abilities in aquatic environments, despite the species not sharing a common ancestor with such traits. Therefore, their flippers are considered **analogous structures**, as they perform the same function but evolve independently.

Hence, the correct answer is option 3, Convergent evolution.

NEET 2023

Ques - 4 Select the correct group/set of Australian Marsupials exhibiting adaptive radiation

Option 1- Tasmanian wolf, Bobcat, Marsupial mole

Option 2 - Numbat, Spotted cuscus, Flying phalanger

Option 3 - Mole, Flying squirrel, Tasmanian tiger cat

Option 4 - Lemur, Anteater, Wolf

Difficulty - Medium

Solution: The **numbat spotted cuscus**, and **flying phalanger** are examples of Australian marsupials that have undergone **adaptive radiation**. This evolutionary process has enabled these species to diversify and adapt to different ecological niches in Australia. The numbat specializes in feeding on termites, the spotted cuscus is an arboreal herbivore, and the flying phalanger has evolved gliding adaptations for movement between trees. Together, they illustrate the remarkable diversity and specialization of marsupials in response to their environment.

Hence, the correct answer is option 2)Numbat, Spotted cuscus, Flying phalanger.

NEET 2022

Ques: 5 - Which of the following statements is not true?

Option 1 - Sweet Potato and Potato is an example of an analogy

Option 2 - Homology indicates common ancestry

Option 3 - Flippers of penguins and dolphins are a pair of homologous organs

Option 4 - Analogous structures are a result of convergent evolution

Difficulty: Medium

Solution: Flippers of Penguins and Dolphins are pairs of analogous organs

Analogous organs have similar functions but dissimilar structures

Hence, Option (3) is correct.

Ques: 6 - Natural selection where more individuals acquire specific character value other than the mean character value, leads to :

Option 1 - Directional Change

Option 2 - Disruptive change

Option 3 - Random change

Option 4 - Stabilising change

Difficulty: Medium

Solution: A Directional Change results from natural selection where more individuals acquire a certain character value that differs from the mean character value.

One extreme phenotype is preferred above others in this kind of selection, which causes the population's features to gradually change in a specific direction. For instance, if birds with larger beaks have an easier time getting food, the population's average beak size may rise over generations, indicating a change in direction.

Hence, the correct answer is option 1)Directional Change.

NEET 2021

Ques: 7 - Ques: 14 - Match **List-I** with **List-II**.

List - I		**List - II**	
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent evolution	(ii)	Bones of forelimbs in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Evolution by anthropogenic action	(iv)	Darwin Finches

Choose the **correct** answer from the options given below.

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Option 1 - **(a)** **(b)** **(c)** **(d)**

(iv) (iii) (ii) (i)

Option 2 - **(a)** **(b)** **(c)** **(d)**

(iii) (ii) (i) (iv)

Option 3 - **(a)** **(b)** **(c)** **(d)**

(ii) (i) (iv) (iii)

Option 4 - **(a)** **(b)** **(c)** **(d)**

(i) (iv) (iii) (ii)

Difficulty - Medium

Solution: (a)Adaptive radiation - Darwin finches (iv): Darwin finches are an example of a species that can evolve into multiple species by adapting to new foods.
(b) Convergent evolution - (iii) Butterfly and Bird Wings: Similar characteristics (wings for flying) are developed by different animals, such as a butterfly and a bird.
(c) Divergent evolution - (ii) Human and whale forelimbs: Although the bones in the forelimbs of humans and whales are similar, their functions differ.
(d) Evolution through human activity - (i) Pesticides and herbicides Resistance: Overuse of chemicals by humans contributes to evolution.

Hence, the correct answer is option 1)(a) - (iv), (b) - (iii), (c) - (ii), (d) - (i).

Ques: 8 - The factor that leads to a founder effect in a population is:

Option 1 - Natural selection

Option 2 - Genetic recombination

Option 3 - Mutation

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Option 4 - Genetic drift

Difficulty: Easy

Solution: Genetic drift is the cause of the founder effect in a population.
When a small number of people from a larger population create a new population, this phenomenon is known as the founder effect. Because of their small size, the new population's genetic diversity is constrained, and some traits might be overrepresented or underrepresented only by accident. This is a type of genetic drift in which tiny populations experience sporadic shifts in allele frequencies, resulting in less genetic diversity.

Hence, the correct answer is option 4)Genetic drift.

NEET 2020

Ques: 9 - Which of the following refers to the correct example(s) of organisms which have evolved due to changes in the environment brought about by anthropogenic action?

A) Darwin’s Finches of Galapagos islands.
B) Herbicide-resistant weeds
C) Drug-resistant eukaryotes
D) Man created breeds of domesticated animals like dogs

Option 1 - only (a)

Option 2 - (a) and (c)

Option 3 - (b),(c) and (d)

Option 4 - only (d)

Difficulty: Easy

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Solution: Anthropogenic actions are due to the direct or indirect effects of human actions on the environment. Resistance due to antibiotics, use of herbicides and artificial breeding of domesticated animals are examples of anthropogenic actions. Deforestation, industrial pollution, and excessive carbon emissions contribute to climate change and habitat destruction. Overfishing and intensive agriculture disrupt natural food chains and reduce genetic diversity in wild populations. These human-induced activities have long-term consequences, affecting both natural ecosystems and human societies.

Hence, the correct answer is option 3 - (b),(c) and (d)

Ques: 10 - Embryological support for evolution was disapproved by

Option 1 - Karl Ernst von Baer

Option 2 - Alfred Wallace

Option 3 - Charles Darwin

Option 4 - Oparin

Difficulty: Medium

Solution: Embryological support for evolution was disapproved by Karl Ernst von Baer. He criticized the idea, originally proposed by Ernst Haeckel, who suggested that embryonic stages reflect an organism's evolutionary history. Ernst von Baer proposed four laws of animal development, which were later referred to as von Baer's laws of embryology.

Hence, the correct option is (1) Karl Ernst von Baer.

Ques: 11 - From the experiments, S.L. Miller produced amino acids by mixing the following in a closed flask:

Option 1 - CH_4, H_2, NH_3, and, water vapor at 800 C°

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Option 2 - CH_3, H_2, NH_4, and, water vapor at 800 Co

Option 3 - CH_4, H_2, NH_3, and, water vapor at 600 Co

Option 4 - CH_3, H_2, NH_3, and, water vapor at 600 Co

Difficulty: Medium

Solution: In 1953, biologist Stanley L. Miller performed a seminal experiment, often referred to as the Miller-Urey Experiment. He aimed to simulate early Earth conditions and explore the potential genesis of organic molecules, particularly amino acids.

Miller's Methodology:

He combined gases mimicking Earth's primordial atmosphere: methane (CH_4), ammonia (NH_4), hydrogen (H_2), and water vapour (H_2O) in a sealed environment.

This gas blend was exposed to electric sparks, replicating lightning's role in the early Earth's energy dynamics.

The system was maintained at an elevated temperature of approximately 800°C for a week.

Results and Significance:

Upon completion, Miller analyzed the contents and observed the formation of various

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organic substances. Crucially, the presence of amino acids is confirmed by this experiment.

These amino acids are essential components of proteins, thus indicating that life's basic building blocks could have emerged under the assumed conditions of our planet's youth.

The Miller-Urey Experiment provided foundational evidence for the abiotic synthesis of organic molecules, offering insight into the origins of life on Earth.

Hence, the correct answer is option 1) CH_4, H_2, NH_3 and, water vapour at 800 Co

Ques: 12 - Flippers of Penguins and Dolphins are examples of:

Option 1 - Adaptive radiation

Option 2 - Convergent evolution

Option 3 - Industrial melanin

Option 4 - Natural selection

Difficulty: Medium

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Solution: Flippers of penguins and dolphins have different structures but similar functions and thus are categorized under analogous organs. Also, penguins and dolphins are not closely related to each other but have evolved similar traits (as per this question - flippers) therefore, show a convergent type of evolution.

hence, the correct option is 2) Convergent evolution

NEET 2019

Ques: 13 - Match the hominid with their **correct** brain size :

(a) Homohabilis (i) 900cc

(b) Homoneanderthalensis (ii)1350 cc

(c) Homo erectus (iii) 650 - 800cc

(d) Homo sapiens (iv) 1400cc

Select the **correct** option.

Option 1 - a-iii, b-i, c-iv, d-ii

Option 2 - a-iii, b-ii, c-i, d-iv

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Option 3 - a-iii, b-iv, c-i, d-ii

Option 4 - a-iv, b-iii, c-i, d-ii

Difficulty: Medium

Solution: Fossils were discovered from the Pleistocene rocks of Olduvai Gorge in East Africa.

(a) Homohabilis had brain capacities between 650 - 800cc.

(b) Homoneanderthalensis had brain capacities of 1400 cc.

(c) Homo erectus had a brain capacity of 900cc.

(d) Homo sapiens had a brain capacity of 1350cc.

Hence, the correct option is (3) a-iii, b-iv, c-i, d-ii

Ques: 14 - Variations caused by mutation, as proposed by Hugo de Vries, are:

Option 1 - Random and directional

Option 2 - Random and directionless

Option 3 - Small and directional

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Option 4 - Small and directionless

Difficulty: Medium

Solution - This theory states that new species arise from the pre-existing one in a single-generation situation by the sudden appearance of marked genetic variation called a mutation. According to De Vries, evolution is a discontinuous and jerky process, rather than a continuous and gradual one. According to Hugo De Vries, mutations are discontinuous, random, and non-directional. These mutations are responsible for the accumulation of variations that act as the raw material for evolution.

Hence, the correct answer is option 2) random and directionless.

Ques: 15 - Analogous structures are a result of:

Option 1 - Divergent evolution

Option 2 - Convergent evolution

Option 3 - Shared ancestry

Option 4 - Stabilizing selection

Difficulty: Medium

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Solution: Analogous organs are those which are similar in function but do not have a common ancestor and thus they are a result of convergent evolution in that different organisms share a common environment. Convergent evolution describes the independent evolution of similar features in species of different lineages. The two species came to the same function, flying, but did so separately from each other.

Hence, the correct answer is Option (2) convergent evolution

NEET 2018

Ques: 16 - The similarity of bone structure in the forelimbs of many vertebrates is an example of

Option 1 - Convergent evolution

Option 2 - Analogy

Option 3 - Homology

Option 4 - Adaptive radiation

Difficulty: Medium

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Solution: **Homologous organs** are structures that may look different and perform different functions but share a similar basic structure, topography, and embryonic origin. Examples include the **forelimbs of vertebrates** like seals (adapted for swimming), bats (adapted for flying), birds (adapted as wings), and humans (adapted for grasping). In these cases, the underlying skeletal structure, such as the bones of the forelimbs, is similar, but they are modified in different ways to suit the specific adaptations of the organism. Homology reflects common ancestry and is a key concept in understanding evolutionary relationships.

Hence the correct answer is option 3, Homology.

Ques: 17 - Among the following sets of examples for divergent evolution, select the **incorrect** option :

Option 1 - Brain of bat, man and cheetah

Option 2 - Heart of Batman and Cheetah

Option 3 - Forelimbs of man, bat and cheetah

Option 4 - Eye of octopus, bat and man

Difficulty: Medium

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Solution: As we learnt in

Types of organic evolution -

1. Divergent Evolution
2. Co-Evaluation
3. Progressive
4. Microevolution
5. Macroevolution
6. Mega evolution
7. Parallel evolution
8. Convergent evolution

Divergent evolution occurs in the same structure, for example - the forelimbs, heart, and brain of vertebrates have developed along different directions due to adaptation to different

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needs whereas the eyes of octopus, bat and man are examples of analogous organs showing convergent evolution.

Hence the correct answer is Option 4. Eye of octopus, bat and man

NEET 2017

Ques: 18 - The first human-like being, the hominid, was called

Option 1 - Homo sapiens

Option 2 - Homo habilis

Option 3 - Homo erectus

Option 4 - Neanderthal man

Difficulty: Medium

Solution: The brain capacity of **Homo habilis**, one of the earliest members of the genus *Homo*, ranged between **650 and 800 cubic centimetres (cc)**. This increase in brain size compared to earlier hominins, such as *Australopithecus*, is associated with improved cognitive abilities, including tool-making and problem-solving skills. *Homo habilis* is

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often referred to as the "handyman" due to its association with simple stone tools, marking an important step in human evolution.

Hence the correct answer is option (2) Homo habilis

Ques: 19 - What was the most significant trend in the evolution of modern man (Homo sapiens) from his ancestors?

Option 1 - Upright posture

Option 2 - Shortening of jaws

Option 3 - Binocular vision

Option 4 - Increasing brain capacity

Difficulty: Medium

Solution: The evolution of modern man (*Homo sapiens*) from his ancestors' "**increasing brain capacity"** is the most significant trend. The human brain is typically 1,330 cm3 in size, more than twice as big as the brains of other primates, including chimpanzees and gorillas. The evolution of modern man (Homo sapiens) from his ancestors' "increasing brain capacity" is the most significant trend. The human brain is typically 1,330 cm^3 in size, more than twice as big as the brains of other primates, including

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chimpanzees and gorillas. This enlargement has facilitated advanced cognitive abilities, such as complex problem-solving, language, and abstract thinking. Over time, the development of the neocortex has allowed humans to adapt to a variety of environments. The increase in brain size is thought to have been driven by both environmental pressures and social complexities.

Hence, the correct answer is option 4 - Increasing brain capacity

Ques: 20 - The process by which organisms with different evolutionary histories evolve similar phenotypic adaptations in response to a common environmental challenge, is called:

Option 1 - Adaptive radiation

Option 2 - Natural selection

Option 3 - Convergent evolution

Option 4 - Non-random evolution

Difficulty: Medium

Solution: Convergent evolution occurs in an unrelated group of organisms. It is the development of similar functional structures but in unrelated groups. Convergent

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evolution occurs when unrelated organisms evolve similar traits independently, often due to adapting to similar environmental pressures or ecological niches. Examples include the wings of bats, birds, and insects, which serve the same function but have different evolutionary origins. This phenomenon highlights the role of natural selection in shaping analogous structures. Convergent evolution demonstrates how similar solutions can arise in different evolutionary lineages to address common challenges.

Hence, the correct answer is option 3) Convergent evolution

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Other Useful Resources

Some useful resources for the preparation of NEET are listed below:

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